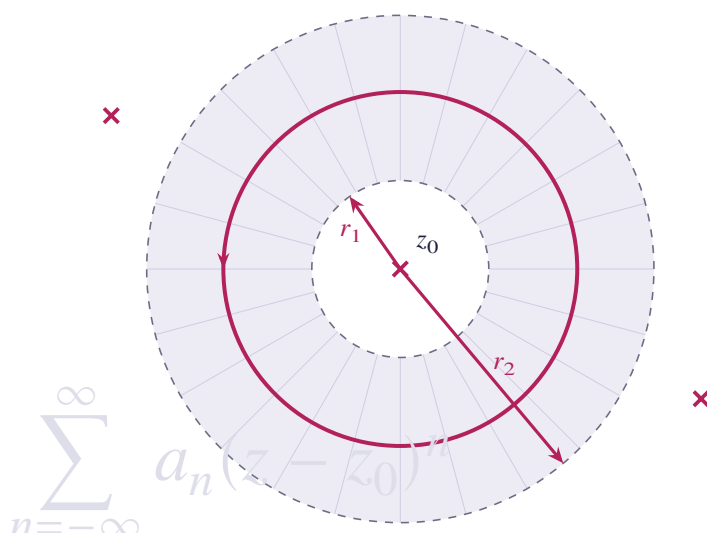


Laurent Series, Singularities and Residues

سری لورنت، نقاط تکین و مانده‌ها



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13.1 The Laurent Series

☺ Theorem 13.1 | Laurent series

If the function $f(z)$ is analytic in the domain D , its Laurent series about the point z_0 is written

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n, \quad a_n = \frac{1}{2\pi i} \oint_C \frac{f(v)}{(v - z_0)^{n+1}} dv$$

where C is a circle centred at z_0 and the function is analytic at all points inside and on this circle, except possibly at z_0 .

The radius of convergence is the distance from z_0 to the nearest non-analytic point of $f(z)$.

For non-negative n we have

$$a_n = \frac{1}{2\pi i} \oint_C \frac{f(v)}{(v - z_0)^{n+1}} dv = \frac{f^{(n)}(z_0)}{n!}, \quad n = 0, 1, 2, \dots$$

Therefore

$$\begin{aligned} f(z) &= \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n + \sum_{n=-\infty}^{-1} a_n (z - z_0)^n \\ &= \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n + \sum_{n=1}^{\infty} a_{-n} (z - z_0)^{-n} \\ &= \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n + \sum_{n=1}^{\infty} \frac{A_n}{(z - z_0)^n}, \quad A_n = a_{-n} \end{aligned}$$

$$A_n = a_{-n} = \frac{1}{2\pi i} \oint_C \frac{f(v)}{(v - z_0)^{-n+1}} dv = \frac{1}{2\pi i} \oint_C (v - z_0)^{n-1} f(v) dv, \quad n = 1, 2, \dots$$

★ Key result 13.2 | The Laurent series: Taylor part and principal part

$$f(z) = \underbrace{\sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n}_{\text{Taylor series part}} + \underbrace{\sum_{n=1}^{\infty} \frac{A_n}{(z - z_0)^n}}_{\text{principal part}}, \quad A_n = \frac{1}{2\pi i} \oint_C (v - z_0)^{n-1} f(v) dv$$

If z_0 is an irregular (singular) point, in other words if the function is not analytic there, the Taylor part alone is not sufficient to describe the function.

■ The Different Cases for the Point z_0

1. If all the A_n are zero, the point z_0 is called a **regular** point. In this case the Laurent series is just the Taylor series.
2. If $A_p \neq 0$ but $(A_n = 0 \ \forall n > p)$, then z_0 is called a **pole of order p** .
3. If z_0 is not a regular point and is also not a pole (in other words, no p can be found such that $A_p \neq 0$ and $(A_n = 0 \ \forall n > p)$), then z_0 is called an **essential singularity**.

13.2 The Residue

Definition 13.3 | Residue

The residue of $f(z)$ at the point z_0 is A_1 .

$$\text{Res}\{f(z)\}\Big|_{z=z_0} = A_1 = \frac{1}{2\pi i} \oint_C f(v) dv$$

Theorem 13.4

If the function $f(z)$ has a pole of order p at z_0 , then

$$A_k = \frac{1}{(p-k)!} \frac{d^{p-k}}{dz^{p-k}} \left[(z-z_0)^p f(z) \right] \Big|_{z=z_0}, \quad k = 1, 2, \dots, p$$

✓ Proof.

$$f(z) = \sum_{n=1}^{\infty} \frac{A_n}{(z-z_0)^n} + \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z-z_0)^n$$

If $f(z)$ has a pole of order p at z_0 we have

$$f(z) = \frac{A_p}{(z-z_0)^p} + \frac{A_{p-1}}{(z-z_0)^{p-1}} + \dots + \frac{A_2}{(z-z_0)^2} + \frac{A_1}{z-z_0} + \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z-z_0)^n$$

Multiplying both sides by $(z-z_0)^p$ gives

$$\begin{aligned} (z-z_0)^p f(z) &= A_p + A_{p-1}(z-z_0) + \dots + A_2(z-z_0)^{p-2} + A_1(z-z_0)^{p-1} + \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z-z_0)^{n+p} \end{aligned}$$

Now if we set $z = z_0$ we get

$$(z-z_0)^p f(z) \Big|_{z=z_0} = A_p$$

If we differentiate $(z-z_0)^p f(z)$ once and then set $z = z_0$:

$$\begin{aligned} \frac{d}{dz} \left[(z-z_0)^p f(z) \right] \Big|_{z=z_0} &= \left\{ A_{p-1} + \dots + A_2(p-2)(z-z_0)^{p-3} + A_1(p-1)(z-z_0)^{p-2} \right. \\ &\quad \left. + \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z-z_0)^{n+p-1} (n+p) \right\} \Big|_{z=z_0} = A_{p-1} \end{aligned}$$

If we differentiate $(z-z_0)^p f(z)$ twice and then set $z = z_0$:

$$\begin{aligned} \frac{d^2}{dz^2} \left[(z-z_0)^p f(z) \right] \Big|_{z=z_0} &= \left\{ 2A_{p-2} + \dots + A_2(p-2)(p-3)(z-z_0)^{p-4} + A_1(p-1)(p-2)(z-z_0)^{p-3} \right. \\ &\quad \left. + \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z-z_0)^{n+p-2} (n+p)(n+p-1) \right\} \Big|_{z=z_0} = 2A_{p-2} \\ \Rightarrow A_{p-2} &= \frac{1}{2!} \frac{d^2}{dz^2} \left[(z-z_0)^p f(z) \right] \Big|_{z=z_0} \end{aligned}$$

■

 Exercise 13.5

Continue the differentiation in the same way and complete the proof of the theorem.

 Remark | Consequence

In particular, when $f(z)$ has a pole of order p at z_0 , the residue is given by

$$\text{Res}\{f(z)\} = A_1 = \frac{1}{(p-1)!} \frac{d^{p-1}}{dz^{p-1}} \left[(z-z_0)^p f(z) \right] \Big|_{z=z_0}$$

13.3 The Residue Theorem

 Theorem 13.6 | The residue theorem

If the function $f(z)$ is analytic everywhere inside the path C except at the points $z_1, z_2, \dots, z_n$, and the residues of the function at these points are $A_1^1, A_1^2, \dots, A_1^n$ respectively, then

$$\oint_C f(z) dz = 2\pi i \sum_{k=1}^n A_1^k$$

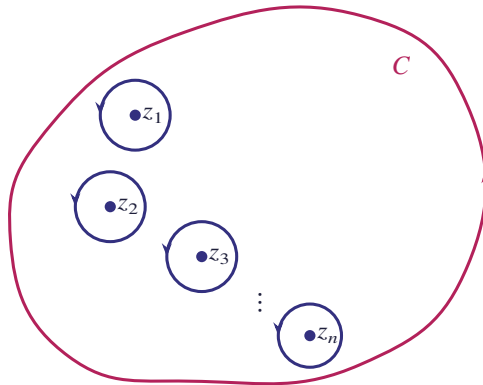


Figure 13.1 A path C and the singular points $z_1, \dots, z_n$ inside it; a small circle is drawn around each point

The proof is left to the reader as an exercise.

13.4 Examples of Taylor and Laurent Expansions

Example 13.7

Find the Laurent series of $f(z) = \frac{1}{z^4 + 1}$ in a neighbourhood of $z_0 = 0$.

We see that $f(z)$ is analytic at $z_0 = 0$, so this is a regular point and the Laurent series is the Taylor (Maclaurin) series. Hence

$$f(z) = f(0) + \frac{1}{1!}f^{(1)}(0)z + \frac{1}{2!}f^{(2)}(0)z^2 + \frac{1}{3!}f^{(3)}(0)z^3 + \dots$$

Geometric series:

$$1 + q + q^2 + q^3 + q^4 + \dots = \frac{1}{1 - q}, \quad |q| < 1$$

$$\frac{1}{1 + z^4} = 1 - z^4 + z^8 - z^{12} + \dots, \quad |z^4| < 1 \Rightarrow |z| < 1$$

(Taylor/Maclaurin series; region of convergence of radius 1.)

$$z^4 + 1 = 0 \Rightarrow z_1 = e^{i\pi/4}, \quad z_2 = e^{i3\pi/4}, \quad z_3 = e^{i5\pi/4}, \quad z_4 = e^{i7\pi/4}$$

We see that the distance from $z_0 = 0$ to the nearest non-analytic point of the function (the radius of convergence) is 1.

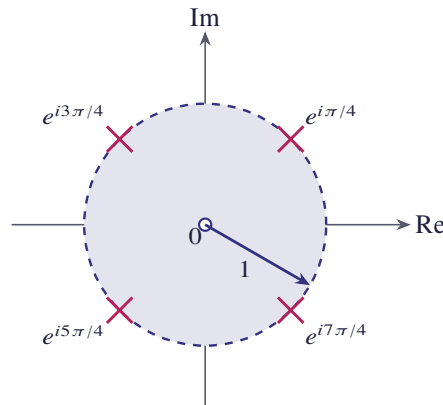


Figure 13.2 The four poles of $z^4 + 1 = 0$ on the unit circle; the radius of convergence of the series about the origin is 1

Example 13.8

Find the Laurent series of $f(z) = \frac{4z^2 + 30z + 68}{(z+4)^2(z-2)}$ in a neighbourhood of $z_0 = 0$.

We see that $f(z)$ is analytic at $z_0 = 0$, so this is a regular point and the Laurent series is the Taylor (Maclaurin) series. Hence

$$f(z) = f(0) + \frac{1}{1!}f^{(1)}(0)z + \frac{1}{2!}f^{(2)}(0)z^2 + \frac{1}{3!}f^{(3)}(0)z^3 + \dots$$

$$\frac{1}{z-2} = \frac{-1}{2-z} = \frac{-1}{2} \frac{1}{1-\frac{z}{2}} = \frac{-1}{2} \left(1 + \frac{z}{2} + \frac{z^2}{4} + \frac{z^3}{8} + \dots \right), \quad \left| \frac{z}{2} \right| < 1 \Rightarrow |z| < 2$$

$$\frac{1}{4+z} = \frac{1}{4} \frac{1}{1+\frac{z}{4}} = \frac{1}{4} \left(1 - \frac{z}{4} + \frac{z^2}{16} - \frac{z^3}{64} + \dots \right), \quad \left| -\frac{z}{4} \right| < 1 \Rightarrow |z| < 4$$

$$\begin{aligned} \frac{d}{dz} \left(\frac{1}{4+z} \right) &= \frac{-1}{(4+z)^2} = \frac{1}{4} \left(-\frac{1}{4} + \frac{2z}{16} - \frac{3z^2}{64} + \dots \right) \Rightarrow \frac{1}{(4+z)^2} \\ &= \frac{1}{4} \left(\frac{1}{4} - \frac{2z}{16} + \frac{3z^2}{64} - \dots \right) \end{aligned}$$

$$f(z) = (4z^2 + 30z + 68) \times \frac{1}{(z+4)^2} \times \frac{1}{z-2}$$

$$f(z) = (4z^2 + 30z + 68) \times \left(\frac{-1}{2} \right) \left(1 + \frac{z}{2} + \frac{z^2}{4} + \frac{z^3}{8} + \dots \right) \times \frac{1}{4} \left(\frac{1}{4} - \frac{z}{8} + \frac{3z^2}{64} - \dots \right)$$

$$f(z) = \frac{-1}{16} \left(34 + 15z + \frac{67}{8}z^2 + \dots \right), \quad |z| < 2$$

(region of convergence of radius 2.)

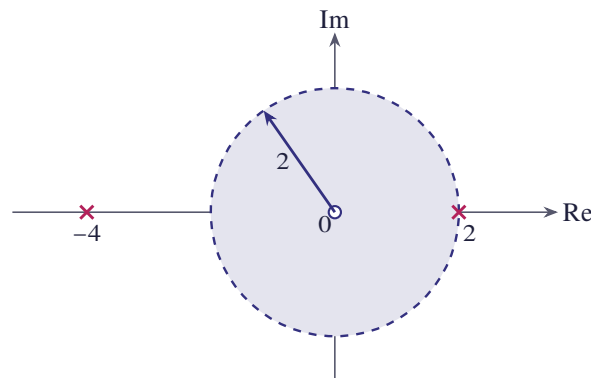


Figure 13.3 The poles $z = -4$ and $z = 2$; the circle of convergence about the origin has radius 2

Example 13.9

Find the Laurent series of $f(z) = \cos z^2$ in a neighbourhood of $z_0 = 0$.

From the definition of the cosine function,

$$\cos z = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \dots$$

$$f(z) = \cos z^2 = 1 - \frac{z^4}{2!} + \frac{z^8}{4!} - \frac{z^{12}}{6!} + \dots$$

The region of convergence is the whole complex plane, with infinite radius of convergence.

Example 13.10

Find the Laurent series of $f(z) = \frac{\cos z}{1 - z^2}$ in a neighbourhood of $z_0 = 0$.

$$\begin{cases} \frac{1}{1 - z^2} = 1 + z^2 + z^4 + z^6 + \dots & |z^2| < 1 \quad |z| < 1 \\ \cos z = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \dots & |z| < \infty \end{cases}$$

$$\begin{aligned} f(z) = \frac{\cos z}{1 - z^2} &= (1 + z^2 + z^4 + z^6 + \dots) \left(1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \dots \right) \\ &= 1 + \left(1 - \frac{1}{2!} \right) z^2 + \left(1 + \frac{1}{4!} - \frac{1}{2!} \right) z^4 + \dots, \quad |z| < 1 \end{aligned}$$

Example 13.11

Find the Laurent series of $f(z) = \frac{1}{z^4}$ in a neighbourhood of $z_0 = 1$.

$$g(z) = \frac{1}{z} = \frac{1}{z - 1 + 1} = \frac{1}{1 + (z - 1)} = 1 - (z - 1) + (z - 1)^2 - (z - 1)^3 + \dots, \quad |z - 1| < 1$$

$$\frac{d}{dz} g(z) = \frac{-1}{z^2} = -1 + 2(z - 1) - 3(z - 1)^2 + 4(z - 1)^3 - \dots, \quad |z - 1| < 1$$

$$\frac{d^2}{dz^2} g(z) = \frac{2}{z^3} = 2 - 6(z - 1) + 12(z - 1)^2 - 20(z - 1)^3 + \dots, \quad |z - 1| < 1$$

$$\frac{d^3}{dz^3} g(z) = \frac{-6}{z^4} = -6 + 24(z - 1) - 60(z - 1)^2 + \dots, \quad |z - 1| < 1$$

$$f(z) = \frac{1}{z^4} = 1 - 4(z - 1) + 10(z - 1)^2 - \dots$$

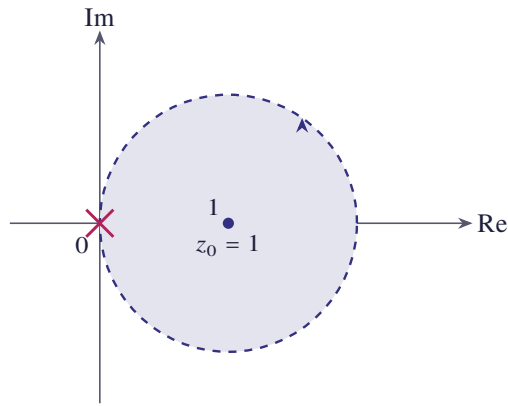


Figure 13.4 The region of convergence of the series about $z_0 = 1$ is a disc of radius 1 (up to the pole $z = 0$)

Exercise 13.12

Find the Laurent expansions of the following functions in a neighbourhood of $z_0 = 0$:

1. $f_1(z) = \frac{1}{\sqrt{1-z^2}}$

2. $f_2(z) = \sin^{-1} z$

3. $f_3(z) = \frac{e^z}{z^2(z^2+1)}$

4. Expand $\sinh z$ in powers of $(z - \pi i)$ and prove that $\lim_{z \rightarrow \pi i} \frac{\sinh z}{z - \pi i} = -1$

5. Prove that $z \cosh z^2 = z + \sum_{n=1}^{\infty} \frac{1}{(2n)!} z^{4n+1}$

13.5 Examples of Singular Points and Residues

Example 13.13

Find the Laurent series of $f(z) = \frac{z}{(z+1)(z+4)^3}$ in the neighbourhoods of its non-analytic points, and find the residue at those points.

The function is not analytic at $z = -1$ and $z = -4$.

(a) In a neighbourhood of $z = -1$, a circle centred at this point with radius less than 3 ($|z + 1| < 3$) does not contain the isolated singular point $z = -4$.

$$\begin{aligned}\frac{z}{(z+4)^3} &= \frac{z+4-4}{(z+4)^3} = \frac{z+4}{(z+4)^3} + \frac{-4}{(z+4)^3} = \frac{1}{(z+4)^2} - \frac{4}{(z+4)^3} \\ \frac{1}{z+4} &= \frac{1}{(z+1)+3} = \frac{1}{3} \frac{1}{1+\frac{z+1}{3}} = \frac{1}{3} \left[1 - \frac{z+1}{3} + \left(\frac{z+1}{3}\right)^2 - \left(\frac{z+1}{3}\right)^3 + \dots \right] \\ \frac{-1}{(z+4)^2} &= \frac{1}{3} \left[-\frac{1}{3} + \frac{2}{3^2}(z+1) - \frac{3}{3^3}(z+1)^2 + \dots \right] \Rightarrow \frac{1}{(z+4)^2} \\ &= \frac{1}{9} - \frac{2}{27}(z+1) + \frac{1}{27}(z+1)^2 - \dots \\ \frac{1}{(z+4)^3} &= \frac{1}{6} \left[\frac{2}{3^2} - \frac{2 \times 3}{3^3}(z+1) + \frac{3 \times 4}{3^4}(z+1)^2 - \frac{4 \times 5}{3^5}(z+1)^3 + \dots \right] \\ &= \frac{1}{27} - \frac{1}{27}(z+1) + \frac{2}{81}(z+1)^2 - \dots \\ \frac{z}{(z+4)^3} &= \frac{1}{(z+4)^2} - \frac{4}{(z+4)^3} = -\frac{1}{27} + \frac{2}{27}(z+1) - \frac{5}{81}(z+1)^2 + \dots \\ f(z) &= \frac{1}{z+1} \frac{z}{(z+4)^3} = -\frac{1}{27} \frac{1}{z+1} + \frac{2}{27} - \frac{5}{81}(z+1) + \dots\end{aligned}$$

Type of singular point: a simple pole (of order 1). Residue: $\text{Res}\{f(z)\}_{z=-1} = -\frac{1}{27}$.

(b) In a neighbourhood of $z = -4$, a circle centred at this point with radius less than 3 ($|z + 4| < 3$) does not contain the isolated singular point $z = -1$.

$$\begin{aligned}\frac{z}{z+1} &= \frac{z+1-1}{z+1} = \frac{z+1}{z+1} - \frac{1}{z+1} = 1 - \frac{1}{z+1} \\ \frac{1}{z+1} &= \frac{1}{z+4-3} = \frac{1}{-3+(z+4)} = \frac{-1}{3} \frac{1}{1-\frac{z+4}{3}} = \frac{-1}{3} \left[1 + \frac{z+4}{3} + \frac{(z+4)^2}{3^2} + \dots \right], \\ |z+4| &< 3 \\ \frac{z}{z+1} &= 1 + \frac{1}{3} + \frac{1}{3^2}(z+4) + \frac{1}{3^3}(z+4)^2 + \frac{1}{3^4}(z+4)^3 + \dots \\ f(z) &= \frac{z}{(z+1)(z+4)^3} = \frac{1}{(z+4)^3} \frac{z}{z+1} \\ &= \frac{\frac{4}{3}}{(z+4)^3} + \frac{1}{3^2} \frac{1}{(z+4)^2} + \frac{1}{3^3} \frac{1}{z+4} + \frac{1}{3^4} + \frac{1}{3^5}(z+4) + \frac{1}{3^6}(z+4)^2 + \dots\end{aligned}$$

Type of singular point: a pole of order 3. Residue: $\text{Res}\{f(z)\}_{z=-4} = \frac{1}{27}$.

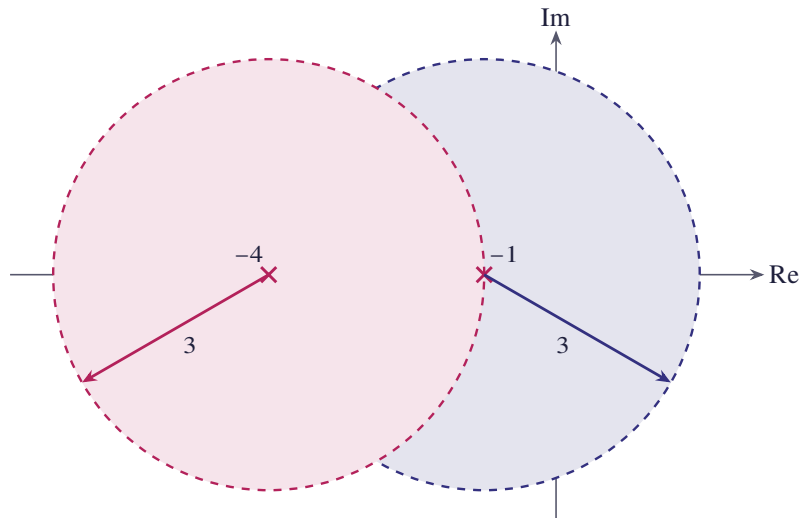


Figure 13.5 The poles $z = -1$ and $z = -4$ of $f(z) = \frac{z}{(z+1)(z+4)^3}$

Example 13.14

Find the Laurent series of $f(z) = e^{1/z}$ in a neighbourhood of its non-analytic point, and find the residue at that point.

$$e^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots \Rightarrow e^{1/z} = 1 + \frac{1}{z} + \frac{1}{2!z^2} + \frac{1}{3!z^3} + \dots$$

Type of singular point: an essential singularity (infinitely many negative powers). Residue: $\text{Res}\{f(z)\}_{z=0} = 1$.

Example 13.15

Find the Laurent series of $f(z) = \cos\left(\frac{z}{z-1}\right)$ in a neighbourhood of its non-analytic point, and find the residue at that point.

$$\frac{z}{z-1} = 1 + \frac{1}{z-1} \Rightarrow \cos\left(\frac{z}{z-1}\right) = \cos\left(1 + \frac{1}{z-1}\right) = \cos 1 \cos\left(\frac{1}{z-1}\right) - \sin 1 \sin\left(\frac{1}{z-1}\right)$$

$$\cos z = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \dots, \quad \sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \dots$$

$$\begin{aligned} \cos\left(\frac{z}{z-1}\right) &= \cos 1 \left(1 - \frac{1}{2!} \frac{1}{(z-1)^2} + \frac{1}{4!} \frac{1}{(z-1)^4} - \dots\right) \\ &\quad - \sin 1 \left(\frac{1}{z-1} - \frac{1}{3!} \frac{1}{(z-1)^3} + \frac{1}{5!} \frac{1}{(z-1)^5} - \dots\right) \\ &= \cos 1 - \sin 1 \frac{1}{z-1} - \cos 1 \frac{1}{2!} \frac{1}{(z-1)^2} + \sin 1 \frac{1}{3!} \frac{1}{(z-1)^3} + \dots \end{aligned}$$

Type of singular point: an essential singularity. Residue: $\text{Res}\{f(z)\}_{z=1} = -\sin 1$.

☹ **Theorem 13.16** | Residue at a simple pole

Prove that if $f(z) = \frac{q(z)}{g(z)}$ and $g(z)$ has a simple root at z_0 , then

$$\text{Res}\{f(z)\}\Big|_{z=z_0} = \frac{q(z_0)}{g'(z_0)}$$

✓ **Proof.**

Since z_0 is a pole of order 1 we have ($p = 1$):

$$\begin{aligned} \text{Res}\{f(z)\}\Big|_{z=z_0} = A_1 &= \frac{1}{(p-1)!} \frac{d^{p-1}}{dz^{p-1}} \left[(z-z_0)^p f(z) \right] \Big|_{z=z_0} = (z-z_0) \frac{q(z)}{g(z)} \Big|_{z=z_0} \\ &= (z-z_0) \frac{q(z)}{g(z) - g(z_0)} \Big|_{z=z_0} = \frac{q(z)}{\frac{g(z) - g(z_0)}{z-z_0}} \Big|_{z=z_0} = \frac{q(z_0)}{g'(z_0)} \end{aligned}$$

✎ **Exercise 13.17**

Find the Laurent series of the following functions about $z = 0$ and determine the type of this point.

$$f_1(z) = \frac{z}{e^z - 1}, \quad f_2(z) = \frac{e^{-z}}{z^3}, \quad f_3(z) = \frac{1}{z^2(1-z^2)}, \quad f_4(z) = \frac{\sin z^2}{z^3}$$